Golla Sathish

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Profile Summary

Aspiring Data Scientist with hands-on experience in Python, SQL, machine learning, deep learning, and NLP through academic and self-driven projects. Skilled in building classification and regression models, data preprocessing, visualization, and model evaluation. Seeking an entry-level Data Science role to apply practical skills to data-driven problem solving.

Technical Skills

Programming: Python, SQL
Data Science: Pandas, NumPy, Data Cleaning, EDA, Data Visualization, Feature Engineering
Machine Learning: Linear/Ridge/Lasso/Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Naive Bayes, XGBoost, LightGBM
Unsupervised ML: K-Means, Hierarchical Clustering, DBSCAN, PCA
Deep Learning: PyTorch, CNN, RNN, Transfer Learning, Computer Vision
NLP: Text Preprocessing, Tokenization, Stopword Removal, Stemming, Lemmatization, TF-IDF
Visualization: Matplotlib, Seaborn
Tools: Jupyter Notebook, VS Code, Streamlit, Git/GitHub

Projects

MNIST Digit Classifier | *PyTorch, CNN, Streamlit* | Live Demo | GitHub *Computer Vision/Deep Learning*

- Built and trained a 2-layer CNN (206,922 parameters) in PyTorch on 60,000 MNIST images using Adam (lr 0.001, batch size 64); training loss fell from 0.23 to 0.02 in 5 epochs.
- Achieved **98.72%** accuracy on the 10,000-image held-out test set.
- Deployed the saved model as a live Streamlit web app where users upload a handwritten digit image and get the predicted digit (0–9).

Student Performance Prediction App | *Scikit-learn, Streamlit* | Live Demo | GitHub *Machine Learning*

- Built a Logistic Regression classifier on 395 student records to predict pass/fail from first- and second-period grades, using a stratified 80/20 split (316 train / 79 test).
- Achieved **87.3%** test accuracy (vs. 67% majority-class baseline) and caught 24 of 26 failing students (92% recall).
- Excluded the final grade to prevent target leakage, and deployed the model as an interactive Streamlit app with sliders for real-time predictions.

RAG with Cross-Encoder Reranking | *LangChain, FAISS, Streamlit* | Live Demo | GitHub *Generative AI / NLP*

- Built a two-stage RAG pipeline with LangChain: split a 720-word document into 18 chunks (500 characters, 100 overlap), embedded with MiniLM-L6-v2, and indexed in FAISS (top-20 retrieval).
- Added a cross-encoder reranker (ms-marco-MiniLM-L-6-v2) that rescores each question–chunk pair and keeps the top 5, cutting the context sent to the LLM by **72%** (5 of 18 chunks).
- Generated answers with gpt-oss-120b (Groq API) from the retrieved context only; deployed as a live Streamlit app.

Sentiment Analysis using TF-IDF and Logistic Regression | *Python, Scikit-learn, Joblib* *NLP / ML*

- Built a system to classify movie reviews as positive or negative, using text preprocessing, TF-IDF vectorization, and Logistic Regression.
- Used a stratified train-test split and evaluated with Accuracy, Precision, Recall, and F1-score.
- Achieved ∼90% accuracy; serialized the vectorizer and model with Joblib.

Education

Sri Venkatesa Perumal College of Engineering and Technology Tirupati, AP
B.Tech in Computer Science and Engineering | CGPA: 7.0/10.0 *Sep 2022 – Sep 2026*

- Relevant Coursework: Python Programming, Data Structures, DBMS, Machine Learning

Sri Lakshmi Narasimha Senior College Anantapur, AP
Intermediate (Class XII) | CGPA: 6.1/10.0 *2020 – 2022*

Additional Information

Languages: English, Telugu (Native), Kannada
Interests: Data Science, Machine Learning, NLP, Problem Solving